# A3-PS003-TC442

## NovaFlow: Bandwidth-Honest Federated Traffic Forecasting

*PS-003 · Federated Traffic-Flow Prediction with Communication-Cost Evaluation*



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# Problem Description

### Bandwidth
Roadside sensors stream raw data to the cloud around the clock over metered cellular links.

### Data concentration
Every junction's travel records pile up in one central store - one breach exposes the whole city.

### Non-IID traffic
A busy highway and a quiet ramp behave differently, so one shared model is pulled in different directions.

**Our question:** *when does federating a traffic model actually pay off - in accuracy, fairness and bytes?*

THE QUESTION NOBODY MEASURES

**≈23×**

more bytes for one 30-round FedAvg training (4.17 MB) than uploading the raw sensor counts it learns from (0.18 MB).

*Prototype measurement · 4 junction-clients · 36 sensors*

# Background

**36**
sensors

**15 min**
sampling interval

**2,101**
time steps

**47**
features per sensor

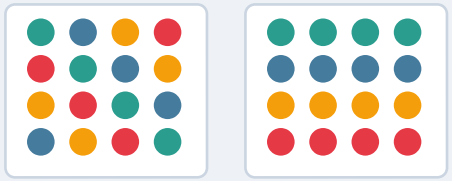
*UCI Traffic Flow Forecasting [7]: two highways, Northern Virginia / Washington D.C., CC BY 4.0*

## Federated learning

Junctions keep their data; only model weights travel. FedAvg averages client models by data size [1]:

$$w = \Sigma_k \, (n_k \, / \, n) \cdot w_k$$

## IID vs non-IID



IID non-IID

Rows = junctions. Non-IID: each junction owns its own roads, which slows and skews FedAvg [2].

## Why bytes matter

- Raw streaming cost grows with time × sensors.
- FL cost grows with model size × rounds × junctions.
- **Which is cheaper depends on the deployment - so it must be measured.**

# Why This Matters for Indian Cities

**ILLUSTRATIVE SCENARIO** One Smart City, four zones: a business district, an IT corridor, a residential belt and a highway bypass. Each zone trains on its own junction data; only model updates reach the traffic command centre.

### Mixed traffic is naturally non-IID

Two-wheelers, autos, buses and cycles share lanes, and the mix changes from zone to zone - exactly the client skew we test.

### Camera junctions flip the cost equation

Where junctions stream video, not counts, raw upload dwarfs model updates (one 2 Mbps camera ≈ 21.6 GB/day).

### Data stays inside each zone

Raw records never leave the zone - in the data-minimisation spirit of India's DPDP Act, 2023 [10] (no compliance claim).

### Honest about the data gap

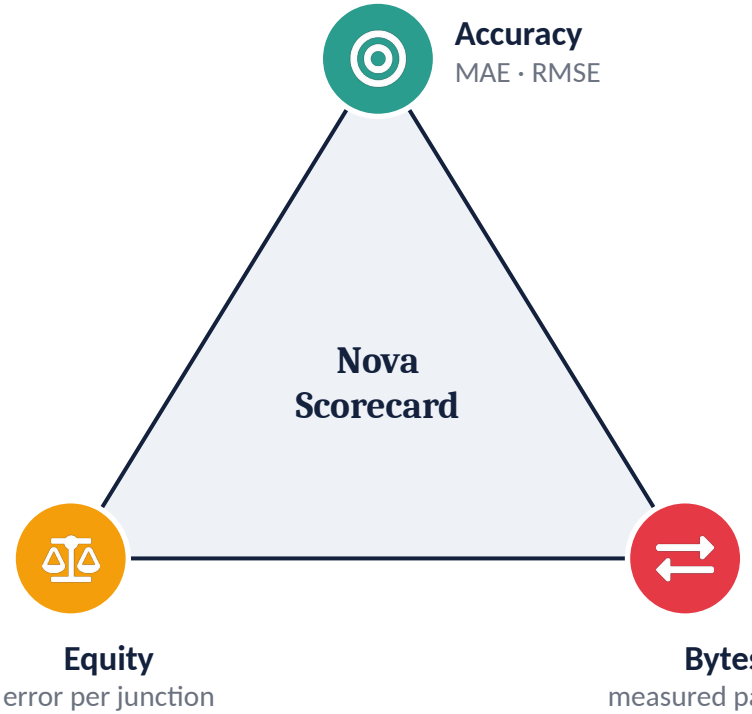
Open Indian traffic datasets are piecemeal [9], so we use a documented public benchmark; the simulator is dataset-independent.

# Literature & Standards Review

| Work | What it contributes | Gap NovaFlow addresses |
|---|---|---|
| **McMahan et al., 2017 [1]** | FedAvg: weighted averaging, far fewer rounds than FedSGD | Benchmarked on images/text, not traffic bytes |
| **Li et al., 2020 - FedProx [3]** | Proximal term stabilises heterogeneous clients | We test it on sensor-grouped (non-IID) traffic |
| **Konečný et al., 2016 [4]** | Quantised / sketched updates cut uplink cost | We measure real serialised bytes, both directions |
| **Liu et al., 2020 - FedGRU [5]** | FL for traffic-flow prediction | Reports accuracy, not a communication break-even |
| **Nishio & Yonetani, 2019 [8]** | Client selection by device resources | PulseFed selects by update size, audited for fairness |
| **IEEE Std 3652.1-2020 [6]** | Architectural framework for federated ML | We map our server / client / comms roles to it |

**Gap:** no study we found reports when FL actually pays off in bytes for scalar traffic sensors - while also checking fairness across junctions.

# NovaFlow: Measure Before You Federate




1. **Audit** verify time order and split before trusting the data
2. **Engineer** 28 features from the last 10 readings + calendar
3. **Partition** 4 junction-clients, IID and non-IID (quiet → busy)
4. **Train** centralised · FedAvg · PulseFed · FedProx · baselines
5. **Meter** serialise and count every byte, up and down
6. **Report** 3 seeds, auto-written report - no hand-typed numbers

# PulseFed: Talk Only When Learning Moves

**Illustrative 6-round schedule**




*16 uploads instead of FedAvg's 24 in this example - a sketch, not a result.*

1. **Score** after local training: r = ‖w_local − w_global‖ / ‖w_global‖
2. **Select** next round, clients whose model moved most talk; at least 2 of 4
3. **Refresh** no client sleeps more than 2 rounds in a row
4. **Aggregate** FedAvg over the active clients, weighted by samples

**Hypothesis:** fewer bytes per round for a small MAE cost.
**Guardrails:** saving capped near 50% (2 of 4 always talk); the Nova equity score checks that quiet roads are not silenced [8].

# Optimizations: Plain FedAvg vs NovaFlow

| Aspect | Before: plain FedAvg | After: NovaFlow |
|---|---|---|
| **Uplink per junction / round** | 17.3 KB float32 weights | **4.8 KB INT8 update (−73%)** |
| **Bytes per round (4 junctions)** | 138.5 KB | **88.3 KB (−36%)** |
| **Feature scaling** | Pool raw rows centrally | **Federated sums only (12.5 KB, once)** |
| **Non-IID drift** | Unhandled | **FedProx (μ = 0.01) compared** |
| **Who talks each round** | All 4 junctions, always | **PulseFed: only movers (min 2)** |
| **Data leakage** | Assumed absent | **Audited by a time-order check** |

*Byte counts measured on our prototype; PulseFed and accuracy effects will be reported after the real-data runs.*

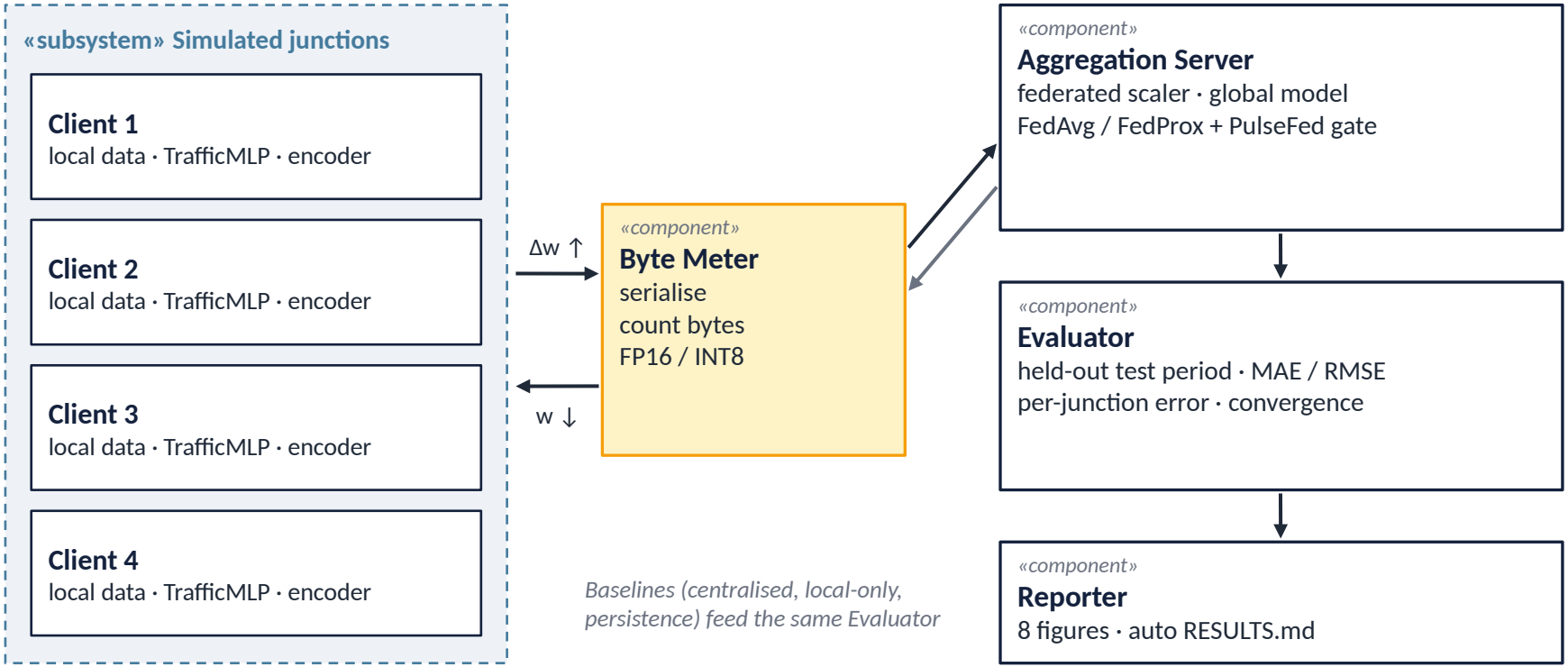
NEW METRIC: COMMUNICATION PAYBACK PERIOD

## ≈300 days

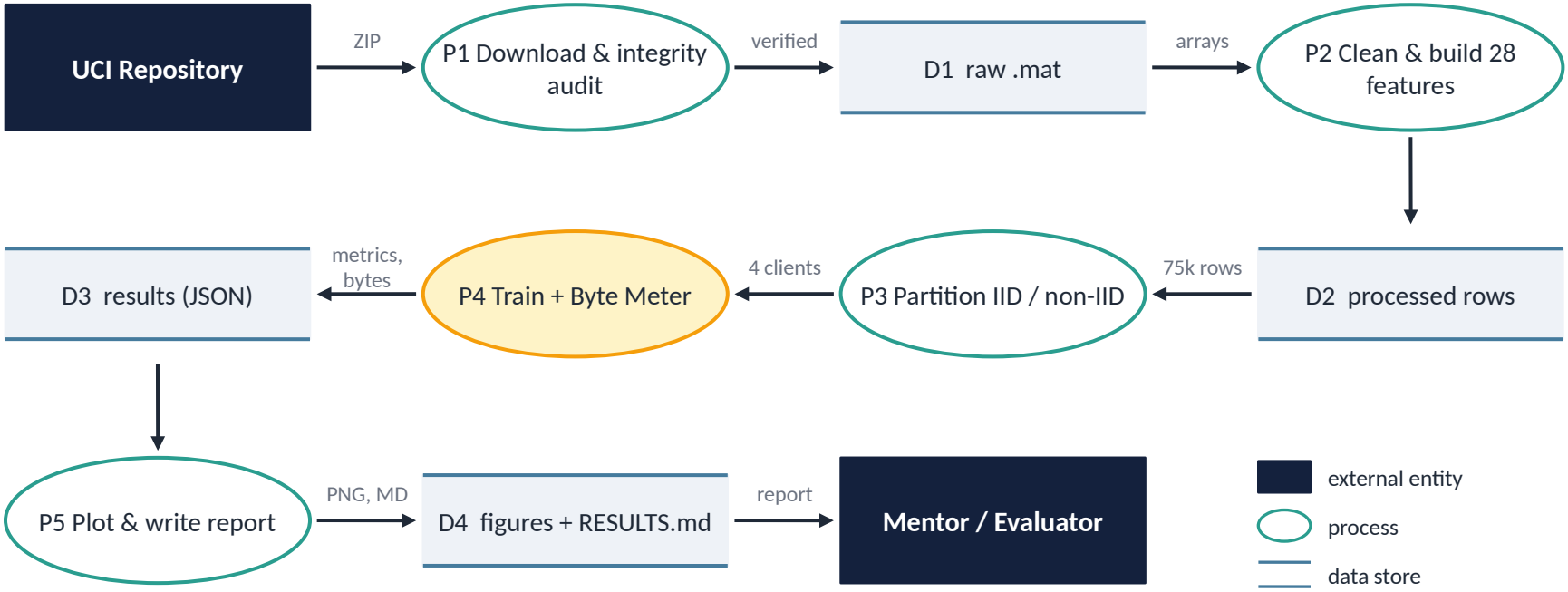
One 30-round training sends as many bytes as streaming all 36 sensors' raw counts for about 300 days.

*For scalar sensors FL must earn its keep through privacy and autonomy - or richer sensors: one 2 Mbps camera ≈ 21.6 GB/day.*

# Solution Architecture & Design



# Data Flow Diagram (Level 1)



# Deployment Timeline

**Oct 13–14**
Level 1: repo, this deck, intro video

**Oct 16–31**
Real-data runs + PulseFed, 3 seeds

**By Oct 31**
Code, report & demo link (Steps 9–10)

**Oct 31–Nov 4**
Evaluator connect & live demo

**Nov 24–30**
Regional hackathon

**Dec 17**
Finals, IIT Jammu

**Status:** prototype simulator built and passing 23 automated tests on synthetic data.

**Test-driven**
23 unit tests

**Continuous checks**
tests run on every push

**Reproducible**
3 fixed seeds, mean ± std

**Zero hand-typed numbers**
report written from results

# Bibliography

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” AISTATS 2017, PMLR 54, pp. 1273–1282.

[2] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, V. Chandra. “Federated Learning with Non-IID Data.” arXiv:1806.00582, 2018.

[3] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith. “Federated Optimization in Heterogeneous Networks.” MLSys 2020.

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[6] IEEE Std 3652.1-2020. “IEEE Guide for Architectural Framework and Application of Federated Machine Learning.” IEEE, 2021.

[7] L. Zhao. “Traffic Flow Forecasting” [Dataset]. UCI Machine Learning Repository, 2019. doi:10.24432/C57897.

[8] T. Nishio, R. Yonetani. “Client Selection for Federated Learning with Heterogeneous Resources in Mobile Edge.” IEEE ICC 2019.

[9] IIT Madras, RBCDSAI. “Synchronized Multi Scale and Multi Sensor Traffic Data From Indian Urban Roads” (project page). rbcdsai.iitm.ac.in.

[10] Government of India. The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023).

# Thank You for Your Attention!

*Measure before you federate.*



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github.com/aiori-hackathon/A3-PS003-TC442